

INTERNAL TEAM HANDOFF · DAY 3 COMPLETE

AI based Network Attack Forecasting from Network Traffic Data

SIH Problem SIH26153 · Dataset CICIDS2017 · 04 September 2026

Final Day 3 Project Progress Report

Ye document public SIH PPT nahi hai. Ye team knowledge + progress handoff hai. Day 2 baseline already lock ho chuka tha. Ye report sirf Day 3 pe **kya verify / complete** hua, uska technical matlab, limitations, aur next stage batati hai.

Member	Role	Day 3
Pranshu	Cybersecurity / Network Analysis / Attack Intelligence	100% DONE
Pulkit	Web Dashboard + Backend/API + System Integration	100% DONE
Pragati	Data Engineering / Dataset Analysis / Temporal Validation	100% DONE
Ankita	ML / Multi-Horizon GRU	COMPLETE
Riddhi	ML Evaluation / Metrics / Early Warning / Comparison	100% DONE
Priyanshi	Python Automation / Reproducibility / Experiment Reliability	100% DONE

TEAM OPERATING NOTE

Day 3 checkpoint ke mutabik currently **sirf Pulkit actively contribute** kar rahe hain, lekin documentation/presentation ke liye original 6-role structure preserve hai. Is report mein kaam role-wise explain kiya gaya hai, taaki koi bhi member apna section judges ko samjha sake.

Repository	SIH26153-Network-Attack-Forecasting
Branch	main · origin/main ke saath up to date
HEAD	897ede8 — Add Priyanshi Day 3 reproducibility audit
Working tree	CLEAN
Report type	Internal Day 3 progress + next-stage roadmap
Language	Simple Hinglish · technical terms ke saath

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1. Is document ka purpose

Day 2 pe model, evaluator, backend foundation, aur data validation lock ho chuke the. Day 3 ka sawal ye tha: kya ye components independently audit + runtime-verify ho sakte hain, bina metrics chhupaye, bina TEST pe tune kiye, bina fake completeness ke?

- Har member Day 3 contribution 30 seconds mein explain kar sake.
- Verified numbers ek hi jagah hon — PPT aur baat match kare.
- COMPLETED / VERIFIED / PARTIAL / LIMITATION / NEXT STAGE mix na hon.
- Day 2 kaam ko Day 3 ka naya result mat samajhna.

2. Executive summary

DAY 3 COMPLETE

Day 3 complete hai. Project ab “model + components exist” se aage badh ke **pipeline, evaluation logic, cybersecurity interpretation, aur real offline demo** independently audit/verify ho chuke hain.

Model sirf ye nahi bata raha ki abhi attack hai ya nahi. Wo last 100 flows dekh kar next 50 / 100 / 200 / 500 flows par future attack-state ke chances predict karta hai. Day 3 ne is story ko MITRE context, H500 ki honest limitation, reproducibility gates, aur browser-verified Attack Forecast demo se joda.

Day 3 pe kya lock hua

- **Pranshu:** MITRE ATT&CK DDoS mapping — T1498 / TA0040; T1498.001 only where behaviour supports; T1498.002 not assigned. File: DOCS/mitre-attack-ddos-mapping.md (commit 2df7d44).
- **Pragati:** Pipeline/tensor integrity re-audit — shapes (13515/2703/2703, 100, 68), 18,921 metadata rows, splits 1–15 / 16–18 / 19–21, 68 features.
- **Priyanshi:** Internal pipeline reproducibility PASS; fresh-clone PARTIAL. File: ml/reports/priyanshi_day3_reproducibility_automation_audit.txt (commit 897ede8).
- **Riddhi:** Evaluator protocol re-confirmed. Older LR vs GRU V1 vs GRU V2 now defensible apples-to-apples (273 windows). Multi-Horizon alag configuration hai — ek table mein merge mat karo.
- **Ankita:** H500 FAR investigation COMPLETE. Operating point 0.35 validation-selected hai. TEST pe retune nahi. H500 exploratory/research horizon.
- **Pulkit:** Real FastAPI + Attack Forecast runtime + npm build PASS + browser localhost:5173/forecast verification.

LIMITATION

H500 FAR = **0.9933**. Ranking signal hai (ROC-AUC 0.7644, PR-AUC 0.8423), lekin current operating point low-FAR alerting ke liye unsuitable hai. Operational narrative H50/H100/H200 prioritize kare.

3. Factual consistency notes

PDF se pehle git, reports, backend, frontend, aur Day 3 checkpoint cross-check kiye. Neeche wala point ek **asli contradiction** hai — silently ek version choose nahi kiya gaya.

3.1 CONTRADICTION — episode attack-start examples

Day 3 prompt ke operational examples ye numbers dete hain: Episode 19 attack **36910** / warning 36709 / lead 201; Episode 20 attack **38910** / warning 38799 / lead 111; Episode 21 attack **40910** / warning 40909 / lead 1.

Wahi prompt ke browser section mein Episode 20 attack start = **39000** likha hai. Repo ki Multi-Horizon early-warning report (`ml/reports/multihorizon_early_warning_report.txt`, commit d3c663e ke baad) verified metadata use karti hai: Episode 19 = **37000**, 20 = **39000**, 21 = **41000**. Backend demo bhi Episode 20 start 39000 se match karta hai.

36910 / 38910 / 40910 **GRU V1** early-warning report mein hain (`ml/reports/gru_early_warning_report.txt`). Ye Multi-Horizon TEST examples nahi hain.

CONTRADICTION — NOT SILENTLY RESOLVED

Dono sources neeche **alag labeled tables** mein hain. Unhe merge mat karo. Multi-Horizon / demo ke liye 37000 / 39000 / 41000 authoritative hain. V1 examples ko MH result mat bolo.

3.2 Jo contradiction nahi, documentation gap hai

- **H500 threshold trade-off TEST numbers** (0.40 / 0.45 / 0.50) dedicated committed report file mein nahi mile. Priyanshi Day 3 audit confirm karta hai ki H500 operating-point alag se investigate hua. Numbers Day 3 checkpoint se hain; repo unhe contradict nahi karta. Source labeled rahega.
- **Riddhi protocol audit** ka alag Day 3 report file git mein nahi hai. LR / GRU V1 / GRU V2 existing reports 273 windows, 30 positive / 243 negative share karte hain. Checkpoint kehta hai protocol ab apples-to-apples verified hai. Dedicated file na hone ko fake file se fill nahi kiya.
- **Pragati Day 3** ka alag naya report file nahi; findings existing tensor/window validation reports + Day 3 checkpoint se match karte hain.

- **requirements.txt / dependency manifest** repo mein nahi hai. Priyanshi audit yahi kehta hai. Fake manifest nahi banaya.

3.3 Jo claim nahi kiya jayega

- /traffic-stream, /alerts, /model-stats real endpoints nahi.
- Poora dashboard real-data driven nahi.
- GRU MITRE ID predict nahi karta.
- Perfect leakage-free preprocessing nahi.
- H500 operationally reliable nahi.
- Live production monitoring nahi — offline demonstrator.
- Is PDF ko already committed mat samajhna jab tak git commit na ho.

4. Day 3 overall status

Member	Day 3 completed scope	Status	Honest leftover
Pranshu	MITRE T1498 mapping document	100% DONE	T1498.001 not universal; T1498.002 not assigned
Pulkit	Runtime + build + browser demo verify	100% DONE	Other SOC pages still mock
Pragati	Pipeline/tensor/split integrity re-audit	100% DONE	Global-median leakage still documented
Ankita	H500 FAR investigation, no cosmetic retrain	COMPLETE	H500 exploratory only
Riddhi	Evaluator protocol + older 273-window comparison	100% DONE	MH vs single-horizon still different setup
Priyanshi	Internal repro PASS; fresh-clone PARTIAL	100% DONE	No deps manifest / runner / README ML section

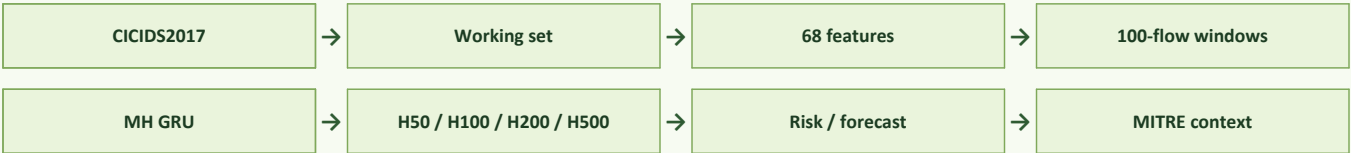
Day 3 workstreams complete/audited. Next stage alag labelled hai.

5. Day 2 baseline vs Day 3 technical progress

Day 2 ne foundation diya. Day 3 ne usi foundation ko audit + interpret + runtime-verify kiya. Restart nahi hua.

Layer	Day 2 baseline (already done)	Day 3 added
Model	MH GRU trained, thresholds locked	H500 operating-point investigation; no retrain
Evaluation	TEST metrics + early-warning evaluator	Protocol audit; LR/V1/V2 273-window comparability
Data	Windows 18 OK; tensors 59 OK	Pipeline/script hard-gate re-audit
Repro	Seed 42 rerun + matching hashes	Internal PASS; fresh-clone PARTIAL documented
Cyber	Attack-pattern narrative; MITRE pending	Verified T1498 mapping document
System	4 real APIs + Attack Forecast wired	Browser runtime + production build verified

End-to-end forecasting pipeline



Evaluation flow — TEST pe threshold tune nahi



Forecasting concept



6. Multi-Horizon TEST classification results

Ye Day 2 se locked TEST numbers hain. Day 3 ne unhe change nahi kiya. Precision/Recall/F1/Accuracy thresholded predictions se. ROC-AUC/PR-AUC probabilities se. FAR = FP / (FP + TN). Thresholds VALIDATION se: 0.30 / 0.55 / 0.45 / 0.35.

Horizon	Precision	Recall	F1	Accuracy	ROC-AUC	PR-AUC	FAR
H50	0.2573	1.0000	0.4093	0.8398	0.9455	0.3681	0.1696
H100	0.4941	0.9833	0.6577	0.8864	0.9492	0.7019	0.1257
H200	0.7572	0.8317	0.7927	0.9034	0.9225	0.7726	0.0761
H500	0.5566	1.0000	0.7151	0.5579	0.7644	0.8423	0.9933

Source: ml/reports/multihorizon_gru_training_report.txt. H500 row highlighted.

Horizon	TN	FP	FN	TP	Positives	Negatives
H50	2120	433	0	150	150	2,553
H100	2101	302	5	295	300	2,403
H200	1943	160	101	499	600	2,103
H500	8	1195	0	1500	1,500	1,203

LIMITATION

H500: 2703 TEST rows; target 0=1203, 1=1500; predicted 0=8, 1=2695; FP=1195; TN=8; TP=1500; FN=0; FAR=0.9933. Recall 1.0 isliye kyunki almost har window ATTACK predict ho rahi hai. Operational alerting ke liye unreliable.

7. Operational early warning

Early warning ka matlab sirf ye nahi hai ki model ne attack ko positive bola. Humein dekhna hota hai ki warning actual attack start hone se **PEHLE** aayi ya nahi. Warning at/after attack start positive-lead nahi maani jaati.

IMPORTANT FINDING

“First warning true-positive forecast thi” aur “warning attack start se pehle aayi” do alag concepts hain. Merge mat karo. MH report: H50 0/3, H100 1/3, H200 1/3, H500 0/3 first warnings also true-positive forecasts.

Horizon	Warning rate	Mean lead	Median lead	≥50 success	≥100 success	FAR
H50	100%	198.7	202	100%	66.7%	0.1696
H100	100%	329.3	189	100%	66.7%	0.1257
H200	100%	341	207	100%	66.7%	0.0761
H500	100%	901	901	100%	100%	0.9933

Sample = 3 TEST episodes only. Lead-time success genuinely early warnings par.

7.1 Multi-Horizon H50 examples (authoritative for MH / demo)

Episode	Attack start	H50 first warning	Lead	H50 probability
19	37000	36704	296	0.372099
20	39000	38798	202	0.321912
21	41000	40902	98	0.324796

Source: ml/reports/multihorizon_early_warning_report.txt. n=3, generalize mat karo.

7.2 GRU V1 examples (alag evaluator — mix na karo)

Episode	Attack start	Warning	Lead
19	36910	36709	201
20	38910	38799	111
21	40910	40909	1

Source: ml/reports/gru_early_warning_report.txt (GRU V1, threshold 0.65). Ye MH numbers nahi hain.

LIMITATION

H500 warning rate 100% aur lead 901 dikh sakta hai, lekin FAR 99.33% ke saath ye operationally unreliable hai. “Best early warning” mat bolo.

8. Baseline / model comparison

Day 3 protocol audit ke baad purana single-horizon comparison defensible hai, kyunki teenon reports same TEST size share karte hain. Multi-Horizon ko isi table mein dump mat karo.

Verified apples-to-apples baseline comparison

Model	F1	ROC-AUC	PR-AUC	FAR	TEST windows
Logistic Regression	0.4091	0.8514	0.3014	0.1646	273
GRU V1	0.6667	0.9444	0.5370	0.1235	273
GRU V2	0.6818	0.9329	0.5420	0.1152	273

Shared protocol (existing reports se verified): 273 TEST windows, 100-flow observation, 68 features, episode-based split, 30 positive / 243 negative. LR = window mean+std + logistic. GRU V1/V2 = sequence GRU, next-100 target. Dedicated Day 3 audit file git mein nahi; numbers inteen reports se hain.

Multi-Horizon GRU horizon-specific evaluation (alag configuration)

Horizon	F1	ROC-AUC	PR-AUC	FAR	TEST windows
H50	0.4093	0.9455	0.3681	0.1696	2,703
H100	0.6577	0.9492	0.7019	0.1257	2,703
H200	0.7927	0.9225	0.7726	0.0761	2,703
H500	0.7151	0.7644	0.8423	0.9933	2,703

PRESENTATION NOTE

PPT mein do tables rakho. “MH H200 F1 0.79 isliye V2 se better hai” — aisa direct rank mat ghoshit karo, kyunki targets/window counts alag hain.

9. H500 investigation (Ankita — Day 3)

Purpose cosmetic TEST metric improve karna nahi tha. Purpose tha: H500 FAR 0.9933 ko samajhna, bina TEST pe threshold tune kiye, bina model retrain kiye.

TEST rows	2,703
Target 0 / 1	1,203 / 1,500
Predicted 0 / 1	8 / 2,695
TN / FP / FN / TP	8 / 1,195 / 0 / 1,500
Selected threshold	0.35 (VALIDATION F1 only)
ROC-AUC / PR-AUC	0.7644 / 0.8423
Current FAR	0.9933

TEST operating-point trade-off (Day 3 investigation; TEST pe selected threshold nahi badla)

Threshold	Precision	Recall	F1	FAR	Note
0.35 (selected)	0.5566	1.0000	0.7151	0.9933	Current locked point
0.40	0.6372	0.8113	0.7138	0.5761	Exploratory only
0.45	0.9372	0.4573	0.6147	0.0382	Exploratory only
0.50	0.9886	0.4053	0.5749	0.0058	Exploratory only

Ye trade-off dikhata hai ki FAR ghatane ke liye recall toot-ta hai. 0.45/0.50 ko naya official threshold mat banao — wo TEST dekh ke choose lagenga. Locked point 0.35 hi hai.

VERIFIED RESULT

H500 mein genuine ranking signal hai, lekin current operating point reliable low-FAR alert ke liye unsuitable hai. Treat as long-range exploratory/research horizon. Operational story: H50/H100/H200.

NEXT PRIORITY

Possible future (labelled): calibration, temporal robustness, improved operating-point selection on VALIDATION. Retrain only with principled methodology. TEST pe tune nahi.

10. Backend + frontend integration

Day 2 ne FastAPI wiring diya. Day 3 ne runtime, TypeScript/Vite production build, aur real page contract verify kiya. Naye fake endpoints nahi add kiye.

API	FastAPI + Uvicorn
Python / Torch	3.13.15 · PyTorch 2.13.0+cpu
FastAPI / Uvicorn	0.141.1 · 0.52.4
Frontend	React / Vite / TypeScript
Inference class	MultiHorizonGRUInference
Checkpoint	ml/models/multihorizon_gru.pt
Runtime	CPU / offline · model.eval() · torch.no_grad()
Input gate	exact 100×68, finite values required

npm run build		PASS (TypeScript PASS, Vite production PASS)
Non-blocking warning		Some chunks > 500 kB after minification — failure nahi
Method	Path	Status
GET	/health	REAL
GET	/model-status	REAL
GET	/demo-sequence	REAL
POST	/predict	REAL
—	/traffic-stream, /alerts, /model-stats	NOT implemented as real APIs

Real frontend/backend flow



Files: backend/main.py, ml/inference/multihorizon_gru_inference.py, website/src/pages/AttackForecast.tsx, website/src/components/forecast/MultiHorizonPredictionCard.tsx, website/src/types/index.ts, website/src/services/api.ts.

11. Browser runtime verification

Real frontend localhost:5173/forecast par open hua. Runtime verification PASS. Ye live packet capture nahi hai — held-out TEST sequence ka offline demo hai.

Field	Verified value
Route	/forecast
Test index / Window	1691 / 17909
Episode	20
Observation flows	38790–38889 (length 100)
Source	CICIDS2017 held-out TEST sequence
H50 / H100 / H200 / H500	97.9% / 99.5% / 99.5% / 98.2%
Predictions	ATTACK / ATTACK / ATTACK / ATTACK
Thresholds	30% / 55% / 45% / 35%
Episode 20 attack start	flow 39000 — window PRE-ATTACK hai

PRESENTATION NOTE

Sequence khud already-attack nahi hai. Attack 39000 par start hota hai; observation 38889 par khatam. Har horizon ka ground-truth target automatically positive nahi. Sahi line: model is pre-attack TEST example par four future horizons ke liye positive attack-state likelihood forecast karta hai.

REAL vs MOCK

Surface	Day 3 status
Attack Forecast + MultiHorizonPredictionCard	REAL model inference
/health, /model-status, /demo-sequence, /predict	REAL FastAPI
Dashboard, LiveTraffic, ThreatIntelligence, ModelPerformance	MOCK / simulation-backed

Surface	Day 3 status
Alerts page	MOCK getAlerts()

12. Cybersecurity / MITRE ATT&CK interpretation

Verified file: `DOCS/mitre-attack-ddos-mapping.md` · commit `2df7d44`. Day 2 pe mapping pending thi. Day 3 pe conservative, evidence-level mapping complete hai.

Primary technique	T1498 — Network Denial of Service
Tactic	Impact / TA0040
Impact type	Availability
T1498.001 Direct Network Flood	Only where observed behaviour supports it
T1498.002 Reflection Amplification	NOT assigned — sufficient evidence nahi
Model relationship	GRU MITRE ID output nahi karta; interpretation layer alag hai

Cybersecurity interpretation flow



VERIFIED RESULT
Sahi narrative: forecasting model future attack-state likelihood predict karta hai; cybersecurity interpretation observed/forecasted behaviour ko ATT&CK context se map karti hai.

WARNING / LIMITATION
Mat bolo: attacker identity, GRU se direct MITRE-ID prediction, universal T1498.001, complete kill-chain visibility.

13. Reproducibility + pipeline audit

Priyanshi: `ml/reports/priyanshi_day3_reproducibility_automation_audit.txt` · commit `897ede8`. Pragati Day 3: usi pipeline ki data/tensor integrity re-audit.

Check	Verified value
Internal pipeline reproducibility	PASS
Fresh-clone reproducibility	PARTIAL / hardening required
X_train / val / test	(13515, 100, 68) / (2703, 100, 68) / (2703, 100, 68)
Dtype	float32, finite
Metadata rows	18,921 (multihorizon_sequence_metadata.csv)
Splits	TRAIN 1–15 · VAL 16–18 · TEST 19–21 · no overlap
Features	68 after removing 10 constant columns
NaN / Inf after prep	0 / 0
Observation length	100 flows
Forecast boundary	forecast_start = observation_end + 1

Hard gates (Priyanshi audit)

- Feature prep: required columns / remaining NaN / Inf → ValueError, phir output.
- Window creation: invalid split, empty output, DDoS-in-observation → ValueError.
- Tensor builder: NaN/Inf, shapes, binary labels, metadata count, forecast boundary, observation length, split/overlap — fail par tensors write nahi.
- Training: tensor shapes, label lengths, test metadata alignment, TRAIN-only weights, VAL-only thresholds, TEST isolation, checkpoint required.

LIMITATION

Fresh clone aaj complete reproduce nahi kar sakta bina extra knowledge ke: no Python dependency manifest, no complete dataset setup guide, no complete README ML reproduction section, no central pipeline runner. Fake requirements.txt is report ke liye nahi banaya.

Preprocessing leakage caveat (Day 2 se retained)

WARNING / LIMITATION

Pipeline mein preprocessing-statistics leakage risk hai kyunki median statistics Inf→NaN ke baad globally, split se pehle compute hote hain. Held-out TEST rows ko imputation ki zaroorat nahi padi, isliye unke existing feature values is issue se directly change nahi hue. Limitation documented hai; future pipeline revision mein harden karna hai. Perfectly leakage-free mat bolo. Sirf is issue ki wajah se retraining automatically required — aisa mat bolo.

14. Important limitations / honest findings

- H500 FAR 0.9933 — operationally unreliable at locked threshold 0.35.
- Early-warning n=3 TEST episodes — large-sample claim nahi.
- MH vs LR/GRU V1/V2 alag evaluation configurations.
- Global median imputation leakage risk (TEST values directly unaltered).
- Fresh-clone reproducibility PARTIAL.
- Dashboard majority still mock.
- Offline demonstrator, live SOC nahi.
- MITRE mapping interpretation layer hai, model output nahi.
- Episode attack-start numbers V1 vs MH mix nahi karne.
- No attacker attribution / complete kill chain.

PRANSHU

Cybersecurity / Network Analysis / Attack Intelligence

Status: Day 3: 100% DONE

15. Pranshu — Day 3 work + next responsibilities

A. Role

Forecast ko defence language dena: DDoS behaviour, attack-stage, MITRE context, detection vs forecasting.

B. Day 3 objective

Day 2 pe mapping pending thi. Day 3 pe evidence-backed MITRE document, bina overclaim ke.

C. Actually complete

DOCS/mitre-attack-ddos-mapping.md, commit 2df7d44. T1498 / TA0040 / Availability. T1498.001 conditional. T1498.002 not assigned. Confirmed vs not-claimed lists. Presentation wording. Predictive cyber-defence takeaway.

D. Simple Hinglish

Model T1498 number nahi thookta. Wo kehta hai: future DDoS-state ke chances kitne hain. Analyst layer kehta hai: ye scenario Network Denial of Service (T1498) ke under aata hai, Impact tactic, availability hit. Direct flood dikhe to T1498.001. Reflection/amplification ka evidence nahi, isliye T1498.002 nahi.

E. Files

- DOCS/mitre-attack-ddos-mapping.md
- DOCS/attack-patterns-and-traffic-signatures.md (Day 2 baseline)
- Git 2df7d44

F–G. Why it matters + remember

Judges ko “F1 0.79” se pehle story chahiye. Yaad: interpretation layer; no attribution; H500 ko MITRE success mat banao; demo pre-attack hai.

H–I. Next

Explainability copy: input pattern → temporal state → probability → horizon → MITRE context. 6-slide PPT ka cyber slide. Expected: 1-page judge script + allowed/not-allowed claims list.

J–K. Q&A

Q1. MITRE mapping ho gayi?

A. Haan, Day 3 document complete hai. Primary T1498. T1498.001 sirf jab behaviour support kare. T1498.002 assigned nahi.

Q2. Model MITRE ID predict karta hai?

A. Nahi. Model future DDoS likelihood. Mapping interpretation layer hai.

Q3. Detection aur forecasting?

A. Detection: abhi attack hai? Forecasting: aage horizon mein attack-state ke chances.

Q4. Attacker kaun hai?

A. Hum attribution nahi karte.

Q5. Kill chain poora dikhta hai?

A. Nahi. Hum DDoS-onset transition tak interpret karte hain.

Q6. Demo par 4/4 ATTACK matlab T1498 confirm?

A. Nahi. Wo forecast probabilities hain, technique classification nahi.

Q7. H500 901 lead ko MITRE early-warning bolo?

A. Nahi. FAR 99.33% hai.

PULKIT

Web Dashboard + Backend/API + System Integration

Status: Day 3: 100% DONE

16. Pulkit — Day 3 work + next responsibilities

A. Role

Trained checkpoint ko demonstrable offline system banana: FastAPI, frontend contract, Attack Forecast page.

B. Day 3 objective

Day 2 wiring ko runtime + production-build + browser se verify karna. Fake completeness nahi.

C. Completed

Four real endpoints confirmed. MultiHorizonGRUInference CPU/eval/no_grad, 100×68 + finite checks, checkpoint thresholds. AttackForecast: getDemoSequence → GET /demo-sequence → predict → POST /predict → card. npm run build PASS; TS PASS; Vite PASS; 500kB chunk warning non-blocking. Browser /forecast: 97.9 / 99.5 / 99.5 / 98.2, all ATTACK, pre-attack episode 20 window.

D. Simple Hinglish

Frontend 100×68 numbers bhejta hai. Backend GRU se 4 probabilities nikalta hai, validation thresholds se 0/1. Demo sequence TEST tensor ka fixed index 1691 hai, isliye offline repeatable hai.

E. Files

- backend/main.py
- ml/inference/multihorizon_gru_inference.py
- website/src/pages/AttackForecast.tsx
- website/src/components/forecast/MultiHorizonPredictionCard.tsx
- website/src/services/api.ts
- website/src/types/index.ts

F–G. Why + remember

Bina verified demo ke judges reports hi dekhenge. Yaad: 4 real APIs only; Attack Forecast real; baaki mock; demo pre-attack; chunk warning failure nahi.

H–I. Next

Decide which mock surfaces actually need real APIs for final demo. Real vs mock labels. Judge flow: health → model-status → demo-sequence → predict → explanation. Deliverable: demo choreography + mock/real map.

J–K. Q&A

Q1. Kaunse real APIs hain?

A. GET /health, GET /model-status, GET /demo-sequence, POST /predict.

Q2. Dashboard live hai?

A. Poora nahi. /forecast real model. Dashboard/LiveTraffic/ModelPerformance mock.

Q3. Build fail to nahi hua?

A. npm run build PASS. 500kB chunk warning non-blocking.

Q4. Demo kahan se aata hai?

A. Held-out TEST index 1691, window 17909, episode 20, flows 38790–38889.

Q5. Ye already attack hai?

A. Nahi. Attack start 39000. Window pre-attack hai.

Q6. GPU chahiye?

A. Nahi. CPU inference.

Q7. Agar 100×68 na ho?

A. POST /predict 422. NaN/Inf reject.

PRAGATI

Data Engineering / Dataset Analysis / Temporal Validation

Status: Day 3: 100% DONE

17. Pragati — Day 3 work + next responsibilities

A. Role

Data ko model-safe temporal windows banana aur leakage/split integrity check karna.

B–C. Day 3

Core scripts re-audited: create_ddos_working_set.py, prepare_ddos_features.py, create_multihorizon_forecasting_windows.py, build_multihorizon_sequence_tensors.py. Tensors float32 finite; metadata 18,921; columns window_id through y500; splits 1–15/16–18/19–21; 68 features; 10 constants removed; NaN=0 Inf=0; forecast starts after observation; length 100.

D. Simple Hinglish

Raw CICIDS ko seedha model mein nahi dalte. Episodes banate hain, 100-flow past window, future horizon targets. Split episode-level hai. Observation ke andar DDoS allowed nahi. Median abhi bhi split se pehle global hai — risk documented, TEST rows impute nahi hue.

E. Files

- ml/scripts/create_ddos_working_set.py, prepare_ddos_features.py
- ml/scripts/create_multihorizon_forecasting_windows.py, build_multihorizon_sequence_tensors.py
- ml/scripts/validate_multihorizon_windows.py, validate_multihorizon_sequence_tensors.py
- ml/reports/multihorizon_validation_report.txt, multihorizon_tensor_validation_report.txt

F–G. Why + remember

Galat split = saare metrics bekaar. Yaad: 42k working set ≠ 5k sample audit; TEST 19–21; leakage caveat; observation DDoS = 0.

H–I. Next

Explainability ke liye 68-feature judge-readable pack. Future: TRAIN-only median design (implement tabhi jab team decide kare). Deliverable: split/leakage one-pager.

J–K. Q&A

Q1. Split kaise hai?

A. Episode-wise. TRAIN 1–15, VAL 16–18, TEST 19–21. Overlap nahi.

Q2. Tensor shape?

A. (N, 100, 68). Train 13515, val/test 2703. float32 finite.

Q3. 68 features kaise?

A. 10 constant columns hataaye. Metadata X mein nahi.

Q4. Leakage?

A. Global median split se pehle. TEST ko impute nahi karna pada. Perfectly leakage-free nahi kahenge.

Q5. Observation mein attack?

A. Validation: observation DDoS = 0. Forecast observation ke baad start.

Q6. Windows kitni?

A. 18,921 metadata rows, 901 per episode × 21.

Q7. dataset_audit_report.txt 42k hai?

A. Nahi. Wo 5,000-row sample audit hai. Mix mat karo.

ANKITA

ML / Multi-Horizon GRU

Status: Day 3: COMPLETE

18. Ankita — Day 3 work + next responsibilities

A. Role

Multi-Horizon GRU owner: shared encoder + four future heads.

B. Day 3 objective

H500 FAR 0.9933 ko investigate karna — cosmetic fix nahi, TEST retune nahi, silent retrain nahi.

C. Completed

Prediction distribution, target-wise probabilities, threshold trade-offs, VAL vs TEST shift implications, episode-wise/operational reading. Locked threshold 0.35. Ranking signal present; operating point poor. Recommendation: H500 exploratory; operational H50/H100/H200.

D. Simple Hinglish

H500 almost saari windows ko ATTACK bol deta hai (2695/2703). Isliye recall 1.0, FAR 0.9933. Agar threshold 0.50 kar do to FAR 0.0058 ho sakta hai lekin recall 0.4053 — aur ye TEST dekh ke choose karna cheating hogi. Isliye 0.35 lock, limitation openly.

E. Files / sources

- ml/scripts/train_multihorizon_gru.py (Day 2 baseline, Day 3 unchanged)
- ml/reports/multihorizon_gru_training_report.txt
- ml/reports/priyanshi_day3_reproducibility_automation_audit.txt §17 H500 note
- Day 3 checkpoint trade-off table (dedicated H500 report file git mein nahi)

F–G. Why + remember

Honesty SIH ka defence hai. Yaad: thresholds 0.30/0.55/0.45/0.35; best epoch 1; H200 balanced classification; H500 exploratory; seed 42 CPU.

H–I. Next

Calibration / VAL-only operating point / temporal robustness — agar team kare to Priyanshi protocol se. Deliverable: written H500 decision already done; next is optional principled experiment, TEST-hidden.

J–K. Q&A

Q1. H500 F1 0.7151 achha nahi?

A. F1 akela nahi. FAR 0.9933 operational fail hai.

Q2. Threshold TEST pe badla?

A. Nahi. 0.35 validation-selected. Trade-off sirf diagnosis.

Q3. Retrain kyun nahi kiya?

A. Cosmetic FAR fix ke liye retrain justified nahi.

Q4. ROC-AUC 0.7644 ka matlab?

A. Ranking signal hai, lekin 0.35 pe almost sab positive.

Q5. Kaunsa horizon demo mein?

A. Operational: H50/H100/H200. H500 research/long-range, limitation ke saath.

Q6. Best epoch 1?

A. Day 2 training: val loss epoch 1 ke baad nahi sudhri. Day 3 ne model change nahi kiya.

Q7. Input?

A. (100, 68) sequence, 4 heads.

RIDDHI

ML Evaluation / Metrics / Early Warning / Comparison

Status: Day 3: 100% DONE

19. Riddhi — Day 3 work + next responsibilities

A. Role

Numbers ki referee: TEST protocol, FAR, early-warning vs classification, fair comparison boundaries.

B–C. Day 3

Evaluator behaviour re-confirmed: TEST 19–21; VAL thresholds reused; no TEST retune; thresholded vs probability metrics split; $FAR = FP / (FP + TN)$; early warning only if first warning before attack start; horizon comparison temporal, not F1 ranking; artifacts unmodified. Older LR/V1/V2: same 273 windows / 30 pos / 243 neg — apples-to-apples under that protocol. MH 2703-window horizon-specific eval alag table.

D. Simple Hinglish

Do alag sawal: (1) is window ka future target positive tha kya? — classification. (2) episode mein pehli warning attack se pehle aayi kya? — early warning. Pehli warning true-positive forecast ho, ye teesra alag count hai.

E. Files

- ml/scripts/evaluate_multihorizon_early_warning.py
- ml/reports/multihorizon_early_warning_report.txt
- ml/reports/forecasting_baseline_report.txt, gru_forecasting_report.txt, gru_forecasting_v2_report.txt
- ml/reports/gru_early_warning_report.txt — V1 examples only

F–G. Why + remember

Overclaim se project tut-ta hai. Yaad: $n=3$; V1 36910 vs MH 37000 mix nahi; two comparison tables; H500 FAR visible.

H–I. Next

PPT appendix: two tables + glossary (FAR, lead, true-positive-first-warning). Deliverable: judge-facing metric legend.

J–K. Q&A

Q1. TEST pe threshold tune?

A. Nahi. 0.30/0.55/0.45/0.35 validation-selected.

Q2. FAR?

A. $FP / (FP + TN)$. H500 = 0.9933.

Q3. 100% warning rate perfect?

A. Nahi. 3/3 episodes pe koi early warning. Quality FAR/precision se.

Q4. V1 201-flow example MH hai?

A. Nahi. Wo GRU V1 report hai. MH episode 19 attack 37000, H50 lead 296.

Q5. LR vs GRU fair?

A. Day 3: haan, 273-window single-horizon protocol. MH usme mix nahi.

Q6. Evaluator ne model change kiya?

A. Nahi. Read-only.

Q7. True-positive first warning vs early?

A. Alag. Early = before attack start. True-positive forecast = us window ka target bhi 1.

PRIYANSHI

Python Automation / Reproducibility / Experiment Reliability

Status: Day 3: 100% DONE

20. Priyanshi — Day 3 work + next responsibilities

A. Role

Experiment kal dubara chale to same result aaye, aur fresh clone ke gaps honestly dikhein.

B–C. Day 3

Full script-by-script audit. Internal pipeline PASS. Fresh-clone PARTIAL. Chain: raw → working set → validate → features → windows → validate → tensors → validate → train → model/preds/report. Hard gates documented. H500 note: TEST pe FAR mat “fix” karo. No fake dependency files.

D. Simple Hinglish

Internal PASS ka matlab: isi machine/workflow par gates aur seed logic strong hain. PARTIAL ka matlab: naya clone dataset, packages, aur command order document ke bina atak sakta hai. Dono status ek saath true hain.

E. File

- ml/reports/priyanshi_day3_reproducibility_automation_audit.txt
- Git 897ede8 (current HEAD)

F–G. Why + remember

Repro judges ko screenshot se pipeline alag karti hai. Yaad: seed 42 Day 2; Day 3 audit PASS/PARTIAL; no requirements.txt yet; don't commit raw CICIDS.

H–I. Next

Hardening (labelled future): dependency manifest, dataset setup doc, README ML section, central runner that orchestrates existing scripts. Optional finite-value check in train script. Deliverable: those docs/runner when team starts Day 4 — not fake-filled now.

J–K. Q&A

Q1. Reproducible hai kya?

A. Internal pipeline: PASS. Fresh clone: PARTIAL.

Q2. Kya missing hai?

A. Dependency manifest, dataset setup guide, README ML section, central runner.

Q3. TEST training mein?

A. Nahi. TRAIN weights, VAL thresholds, TEST last.

Q4. H500 FAR script se fix?

A. Audit mana karta hai TEST pe threshold change karke.

Q5. Tensors fail hon to?

A. Builder RuntimeError, files write nahi.

Q6. Raw data git mein?

A. Nahi. Intentionally gitignored.

Q7. Day 2 hash rerun?

A. Day 2 PASS. Day 3 ne cosmetic retrain nahi kiya.

21. Next stage / Day 4 roadmap

Ye Day 3 future work nahi hai. Day 3 complete hai. Neeche continuation priorities hain.

#	Priority	Who	Output
1	Explainability (input → state → forecast → MITRE)	Pranshu + Pragati + Pulkit UI if shown	Judge-friendly explanation, no fake SHAP claims
2	H500: calibration / VAL operating point / shift	Ankita + Riddhi; Priyanshi if rerun	No TEST tune; optional principled experiment
3	Backend only if demo needs it	Pulkit	traffic/alerts/stats — only if required
4	Dashboard hardening, real vs mock labels	Pulkit	Critical mock replace gradually
5	Offline demo choreography	Pulkit + Priyanshi + Pranshu copy	Deterministic judge flow
6	Final SIH PPT — exactly 6 slides	All six roles	Simple story + honest limits
7	Presentation rehearsal	All	Each person: work, why, limit, next

NEXT PRIORITY

Pehle explainability + 6-slide PPT. H500 ko chhupa ke PPT mat banao.

22. SIH PPT / presentation preparation

Exactly 6 slides. Technical dump nahi. Offline demonstrator honesty ke saath.

- 1. Problem: late detection vs forecasting need
- 2. Idea: 100 flows → H50/H100/H200/H500
- 3. Evidence: episode split, TEST 19–21, H200 balanced, H500 FAR 0.9933 limitation
- 4. System: real /forecast demo, 4 APIs, pre-attack TEST window
- 5. Cyber: T1498 interpretation layer, not model output
- 6. Limits + next: mock pages, leakage caveat, explainability

Pranshu	Forecast → T1498 context. Model MITRE ID nahi bolta.
Pulkit	4 real APIs, /forecast live, baaki mock, demo pre-attack.
Pragati	Episode split, 100×68, leakage caveat, 0 observation DDoS.
Ankita	4-head GRU, VAL thresholds, H500 exploratory.
Riddhi	TEST protocol, FAR, early ≠ F1, two comparison tables.
Priyanshi	Internal PASS, fresh-clone PARTIAL, seed/gates.

PRESENTATION NOTE

PPT mein mat likho: live production, perfectly leakage-free, all pages real, H500 best, GRU predicts T1498, attacker ID, 273-vs-2703 as one ranking.

23. Final project status + next milestone

Day 3 milestone **achieved**: audited pipeline + honest H500 + MITRE interpretation + real offline browser demo + clean git checkpoint 897ede8. Next milestone: 6-slide PPT + explainability + demo choreography.

Layer	Day 3 state	Next milestone
Cyber	T1498 mapping documented	Judge script + explainability language
Data	Pipeline/tensor gates re-audited	Optional TRAIN-only median design
Model	H500 investigated, not cosmetically fixed	Optional VAL-only calibration
Eval	Protocol + 273-window baseline labelled	Two-table PPT appendix
Repro	Internal PASS / clone PARTIAL	Manifest + README + runner
System	Browser-verified /forecast	Mock/real hardening as needed

TEAM BOTTOM LINE

Team bottom line: model future horizons forecast karta hai; H200 currently sabse balanced TEST classification; H500 FAR 0.9933 ke bina mat present karo; Attack Forecast real hai, poora dashboard nahi; MITRE interpretation hai, model ID nahi; TEST episodes 19–21; demo window pre-attack (episode 20 start 39000).

Appendix — locked facts

HEAD	897ede8
MITRE file	DOCS/mitre-attack-ddos-mapping.md
Priyanshi audit	ml/reports/priyanshi_day3_reproducibility_automation_audit.txt
Thresholds	H50 0.30 · H100 0.55 · H200 0.45 · H500 0.35
H200 F1 / FAR	0.7927 / 0.0761
H500 FAR	0.9933
Real APIs	/health · /model-status · /demo-sequence · /predict
Browser demo	H50 97.9% · H100 99.5% · H200 99.5% · H500 98.2% · all ATTACK
This PDF source	DOCS/day3_progress_report/generate_day3_progress_report.py

End of Day 3 internal progress report. Next stage isi “next” column se start hoga.